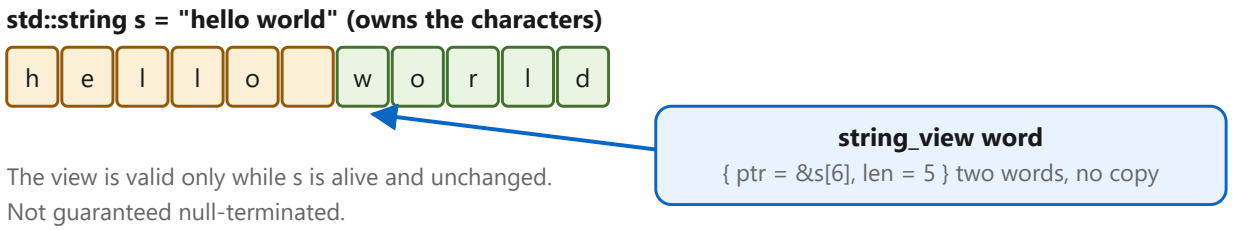
Strings, Regex and Time

Text in C++: ownership with `string` and `string_view`, how a string stores its characters, converting numbers, formatting and streams, then `<regex>` and `<chrono>`.

`std::string` and `std::string_view`



View for read-only parameters; string for anything you store.

`std::string` **owns** its characters. `std::string_view` (C++17) is a **non-owning** pointer plus length: two words, no allocation, valid only while the text lives.

```
#include <string>
#include <string_view>

void log(std::string_view msg);          // accepts std::string, string
literal, char array ...  without copying

std::string s = "hello world";
log(s);
log("literal");                          // no temporary std::string is built

auto pos = s.find("world");              // 6
if (pos != std::string::npos) {          // find returns npos (a huge value)
when nothing matches
    std::string_view word = std::string_view(s).substr(pos);   // "world", no
copy
}
```

CLASSIC BUG: DANGLING VIEW

```
std::string_view bad = std::string("temporary");   // the string dies
at the semicolon
// 'bad' now points at freed memory: undefined behaviour if you read it

std::string_view f() {
    std::string local = "oops";
    return local;                                   // returns a view
of a dead local
}
```

A view must not outlive its string and is **not null-terminated**: do not pass `sv.data()` to a C API.

- **SSO:** short strings (about 15 chars) live inside the object, no heap allocation.
- `string_view` for read-only parameters; `std::string` for what you store.
- Building big strings in a loop: `append` or `reserve`, not chained `+` temporaries.

KEY POINT View when you only read (accepts literals without a temporary); `const string&` when you need null termination.

Numbers and text: `from_chars`, `to_chars`, `stoi`

FUNCTION	FAILURE	ALLOCATES?	LOCALE?	USE
`std::from_chars` / `to_chars` (C++17)	returns an error code, **no exception**	no	no	**hot paths**, parsing feeds
`std::stoi`, `stod`, `to_string`	throws `invalid_argument` / `out_of_range`	yes (a `string`)	yes (slower)	ordinary code
`atoi`, `strtol`, `sscanf`	silent 0 or `errno`	no	yes	avoid in new C++
`std::istringstream >>`	sets `failbit`	yes	yes	mixed text parsing

```
int v = 0;
std::string_view num = "1234";
auto [ptr, ec] = std::from_chars(num.data(), num.data() + num.size(), v);
if (ec == std::errc{} && ptr == num.data() + num.size()) use(v);      //
parsed the WHOLE input

char buf[32];
auto r = std::to_chars(buf, buf + sizeof buf, 255, 16);                //
"ff": no allocation
std::string_view out(buf, r.ptr - buf);
```

- Compiled: `from_chars("12x4")` parsed **12**, reported no error, and left `ptr` at `"x4"`. **Always check that `ptr` reached the end** when the whole token must be a number.
- Compiled: `stoi("abc")` threw `invalid_argument` and `stoi("99999999999")` threw `out_of_range`.
- `from_chars` does not skip leading whitespace and does not accept a leading `+`.
- Floating-point `from_chars` needs a recent standard library (GCC 11+, MSVC 2019+).

Formatting: `std::format` and streams

```cpp
#include <format>                                   // C++20

std::format("{:>8}|{:<8}|{:^8}|", "right", "left", "mid");   // "
right|left    |  mid   |"
std::format("{:08.3f} {:#x} {:#b} {:+d} {}", 3.14159, 255, 5, 7, true);   //
"0003.142 0xff 0b101 +7 true"
std::format("{1} before {0}", "world", "hello");            // "hello before
world": positional arguments
```

SPEC	MEANING
`{:>8}` `{:<8}` `{:^8}`	right, left, centre in a field of width 8
`{:08.3f}`	zero padded, width 8, 3 digits after the point
`{:#x}` `{:#b}`	hex and binary with the `0x` / `0b` prefix
`{:+d}`	always show the sign

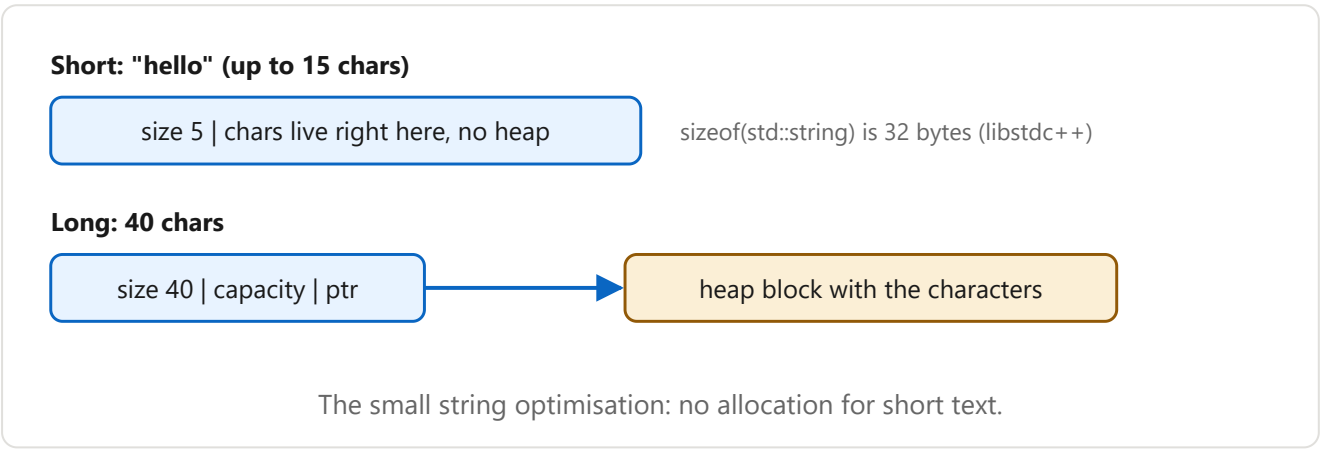
Which one to use

TOOL	GOOD	BAD
`std::format` (C++20)	type-safe, readable, fast, no state, errors at compile time	allocates a `std::string`; check your compiler supports it (GCC 13+)
`std::print` (C++23)	`format` straight to `stdout`	C++23
`operator<<` streams	works everywhere, extensible for your types	stateful (`std::hex` sticks!), verbose, slower
`printf`	fast, familiar	no type checking: a wrong specifier is undefined behaviour

> **STREAMS KEEP STATE** `std::cout << std::hex << 255;` leaves every later integer printed in hex until you switch back with `std::dec`. `setw` applies only to the next output. `std::format` has none of this.

- To format your own type, specialise `std::formatter<T>` (or provide `operator<<` for streams).
- `std::format_to(out, ...)` writes into an existing buffer or iterator to avoid the temporary string.
- Do not build log strings on a hot path unless the log level is enabled.

Inside `std::string`: SSO and capacity



The small string optimisation: no allocation for short text.

A `std::string` is a small object (compiled: **32 bytes**, libstdc++) that holds a pointer, a size and a capacity. For short text it stores the characters **inside the object**, so there is no heap allocation: **SSO**. Compiled: an empty string has capacity **15**, `"hello"` was stored inside the object and a 40-character string was not.

FACT	WHY IT MATTERS
SSO limit is implementation-defined (15 in libstdc++, 22 in libc++)	do not rely on the exact number
Capacity is the room before the next reallocation; **size** is the length	`s.size() ≤ s.capacity()`
Growth is geometric (usually doubling)	compiled: appending 1000 characters one at a time reallocated **7 times**
`reserve(n)` allocates once up front	compiled: with `reserve(1000)` first, the buffer never moved
A reallocation invalidates pointers, references and iterators into the string	do not keep `s.c_str()` across a modification
Moving a long string steals the pointer; moving a short one copies the inline bytes	a move is cheap but never worse than a copy

```
std::string s;
s.reserve(1000);                        // one allocation, then no
reallocations while we append up to 1000
for (int i = 0; i < 1000; ++i) s += 'a';

std::string t = "a" + name + "b";     // each + may build a temporary: prefer
append, a format call or an ostringstream
```

KEY POINT "What is SSO?" Short strings are stored in the string object, so no heap allocation and no pointer chase. It is why short-string copies are cheap and why `string` moves are not always pointer steals. On a hot path `reserve` and avoid building strings at all (Performance & Memory).

String streams and files

TYPE	READS FROM / WRITES TO	HEADER
`std::istringstream`	reads from a `std::string`	`<sstream>`
`std::ostringstream`	builds a `std::string` (`.str()`)	`<sstream>`
`std::ifstream` / `ofstream` / `fstream`	a file	`<fstream>`

```
std::istringstream in("42 3.5 word");
int a; double b; std::string w;
in >> a >> b >> w;                          // 42, 3.5, "word": >> skips whitespace and converts

std::ostringstream out;
out << "id=" << 7 << " px=" << 101.25;
std::string s = out.str();                  // "id=7 px=101.25"

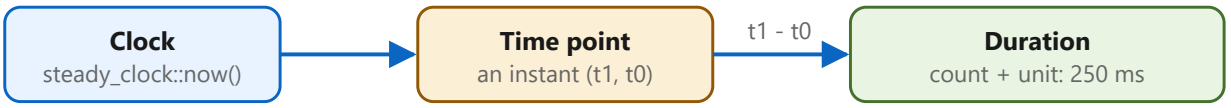
{
    std::ofstream f("data.txt");            // opened by the constructor, CLOSED by the destructor (RAII)
    f << "first line\nsecond line\n";
}
std::ifstream f("data.txt");
std::string line;
while (std::getline(f, line)) { /* one line at a time */ }     // the loop condition tests the read itself

std::istringstream csv("apple,5000.25,10");
for (std::string field; std::getline(csv, field, ',');) { /* "apple", "5000.25", "10" */ }
```

- **Loop on the read, not on `eof()`:** `while (std::getline(f, line))` is right; `while (!f.eof())` processes the last line twice.
- Check the stream after opening: `if (!f) { /* could not open */ }`. A failed stream silently does nothing afterwards.
- Stream state bits: `good`, `eof`, `fail` (a format error, such as reading text into an `int`), `bad` (a hard I/O error). `clear()` resets them.

- Compiled: `getline(csv, field, ',')` split a three-field line, and a file of two lines read back as two lines.
- Files are closed by the destructor, so a plain scope is the RAII way to flush and close.
- `std::endl` flushes every time (slow). Use `'\n'`.

Time with `std::chrono`



Elapsed time and timeouts: steady_clock (monotonic). Timestamps for people: system_clock (can jump).

Durations are type-safe: seconds and milliseconds cannot be confused.

Three ideas, in this order:

- **Duration**: count plus unit; type-safe.
- **Time point**: an instant since a clock's epoch.
- **Clock**: the source of time points.

CLOCK	BEHAVIOUR	USE IT FOR
`system_clock`	Wall-clock time. Can jump (NTP, manual changes, daylight handling by the OS). Convertible to `time_t`.	Timestamps, dates shown to people
`steady_clock`	**Monotonic**: never goes backwards.	Measuring elapsed time, timeouts
`high_resolution_clock`	Smallest tick the implementation has. Often just an alias of one of the above.	Rarely the right choice; prefer `steady_clock`

```
#include <chrono>
#include <thread>
using namespace std::chrono_literals;

auto t0 = std::chrono::steady_clock::now();
std::this_thread::sleep_for(250ms);                    // duration literal
```

```cpp
auto t1 = std::chrono::steady_clock::now();

auto elapsed = std::chrono::duration_cast<std::chrono::milliseconds>(t1 - t0);
std::cout << elapsed.count() << " ms\n";                    // about 250

std::chrono::seconds s = 90s;
std::chrono::milliseconds ms = s;                            // implicit: lossless conversion allowed
// std::chrono::seconds back = ms;                           // ERROR: would lose precision, needs duration_cast
auto m = std::chrono::duration_cast<std::chrono::minutes>(s);   // 1 (truncates)
```

> **STEADY_CLOCK FOR INTERVALS** `system_clock` can jump, giving negative or wrong intervals: bad for timeouts.

Regular expressions

A regex describes a *set of strings*. Match vs search is a favourite question:

FUNCTION	QUESTION IT ANSWERS
`std::regex_match`	Does the **whole** string match the pattern?
`std::regex_search`	Does **some part** of the string match? (first match; captures in `std::smatch`)
`std::regex_replace`	Replace every match with a replacement string.

```cpp
#include <regex>

std::regex re("ab+c");
std::regex_match("abbbc", re);       // true  : the entire string fits
std::regex_match("xabbbcx", re);     // false : extra characters around it
std::regex_search("xabbbcx", re);    // true  : found inside
```

SYNTAX	MEANING	EXAMPLE
`.`	any character	`a.b` matches `aab`, `axb`
`?` `*` `+`	0 or 1, 0 or more, 1 or more	`ab*c` matches `ac`, `abbbc`
`{n}` `{n,m}`	exactly n, between n and m	`ab{3}c` matches `abbbc`
`[abc]` `[a-z]` `[^a-z]`	one of, a range, none of	`a[bcd]e`
`(x|y)`	group and alternation	`gr(e|a)y` matches `grey`, `gray`
`^` `$`	start and end of the string	`^\d+$`
`\d` `\w` `\s`	digit, word character, whitespace	`\w+@\w+`

`std::regex` uses the ECMAScript grammar.

> **BACKSLASHES** Write `"\\d+"` or a raw string `R"(\d+)"`. POSIX classes like `[[:alnum:]]` work in `std::regex`.

Captures, replace and iterating matches

Parentheses make **capture groups**; `std::smatch` holds the whole match at 0, groups after.

```
std::string line = "temp=21.5C humidity=40%";
std::regex kv(R"((\w+)=([\d.]+))");

std::smatch m;
if (std::regex_search(line, m, kv)) {
    // m[0] = "temp=21.5"   m[1] = "temp"   m[2] = "21.5"
    std::cout << m[1] << " is " << m[2] << '\n';
}

// every match, not just the first
for (std::sregex_iterator it(line.begin(), line.end(), kv), end; it != end;
++it)
    std::cout << (*it)[1] << " -> " << (*it)[2] << '\n';   // temp -> 21.5,
then humidity -> 40
```

```
// replace: $1, $2 refer to groups
std::string swapped = std::regex_replace(line, kv, "$2:$1");   // "21.5:tempC
40:humidity%"  (only the matched parts change)
```

PRACTICAL LIMITS OF STD::REGEX

- **Constructing a regex is expensive**: build once, not in a loop.
- `std::regex` is **slow**; on hot paths use `find` / `from_chars` (or RE2, CTRE).
- Nested quantifiers like `(a+)+b` can backtrack exponentially: no untrusted input.
- Invalid pattern: `std::regex_error`.

KEY POINT `regex_match("hello world", regex("hello"))` is false: it must cover the **entire** string. `regex_search` is true.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
std::string and std::string_view	C++ Primer	9.5 Additional string operations	389 to 396
String streams and files	C++ Primer	8.1 to 8.3 IO classes, file IO and string streams	339 to 353
	Programming: Principles and Practice	9.11 String streams	296 to 297
Regular expressions	C++ Primer	17.3 Regular expressions	757 to 773